

Rev A (corrected) · Board outline 21.0 x 17.5 mm, XIAO-compatible · Sheet 1 of 1

Rev A (corrected) · Board outline 21.0 x 17.5 mm, XIAO-compatible · Sheet 1 of 1

42 — to doorlock MCU
(Tuya keep-alive header)

3 SRDY

4 MRDY

1 TXD

2 RXD

arrows show
signal direction

Protocol
UART 115200 8N1
Tuya framing, 0x55AA header

Keep-alive: module always powered
SRDY / MRDY gate the UART traffic

TWT enabled
1.4+ years on 2400mAh, per Tuya spec

GPIO8 / GPIO9 are the actual ESP32-C6 strap pins - not GPIO0. SW1 now resets via EN and SW2 boots via GPIO9. GPIO0 is a normal free GPIO on this chip.

J2 carries 4 signals only. Confirm chassis or wiring harness ties this board's GND to the lock MCU's GND, or add a 5th pin.

 chassis / signal ground

+3.3V +3.3V rail tap

net label (same color+name = same net)

signal direction

21.0 x 17.5mm is very tight for 3 connectors, 2 switches and an LDO. Use low-profile SMD headers/switches and consider double-sided placement (module top side, passives bottom) before finalizing the fab footprint library.

Reset - pulls EN low

SW2



GPIO9 / BOOT

Boot - pulls GPIO9 low

EN / GPIO9 / TXD / RXD tags connect to the matching tags of the same color and name in the Core Module section (net labels, not drawn as long wires)

ESP32-C6-MINI-1-N8 Doorlock Baseboard — Engineering Specification

Rev A (corrected) — Tuya Keep-Alive Wi-Fi adapter module · companion sheet: schematic (page 1)

1. Overview

This board replaces the Wi-Fi module in a Tuya-compatible smart doorlock. It carries its own battery-fed 3.3V supply (the lock’s MCU cannot source Wi-Fi peak currents), and stays powered continuously — the **Keep-Alive** topology — with the UART link gated by a hardware handshake pair (SRDY/MRDY) rather than by powering the module on and off.

- **Core:** ESP32-C6-MINI-1-N8 (ESP32-C6FH8, RISC-V @160MHz, 8MB flash, PCB antenna, integrated matching network)
- **Protocol:** UART 115200 8N1, Tuya framing (0x55AA header)
- **Power:** single-cell-equivalent battery input, 3.6-5.5V, reverse- and transient-protected, regulated to 3.3V
- **Expected battery life:** 1.4+ years on 2400mAh with TWT enabled, per typical Tuya module figures — validate against your actual duty cycle before it goes on packaging.

2. Design review notes — please read before layout

I corrected two things in your original component list before drawing the schematic. Both are called out on the schematic sheet in red as well.

2.1 Strapping pins — GPIO0 is not one of them on ESP32-C6

On the classic ESP32, GPIO0 is the boot-select strap. On **ESP32-C6 that job moved**: the real strapping pins are **GPIO8 and GPIO9** (plus the JTAG-alias pins MTMS/MTDI, not relevant here). GPIO0 is a normal, unremarkable GPIO on this chip.

Practical effect on your original spec: the **“Reset” button on GPIO9** would not have reset anything — GPIO9 isn’t wired to the reset logic, and if it happened to be low at power-up for any other reason the module would boot into UART download mode instead of running your application. The **“Boot” button on GPIO0** would not have entered download mode, since GPIO0 has no strapping function on this chip.

Function	Old (as specified)	Corrected
Reset button	GPIO9 → GND	EN (CHIP_PU) → GND
Boot/download button	GPIO0 → GND	GPIO9 → GND
R2 pull-up	GPIO0 → 3.3V	GPIO9 → 3.3V (real strap; needs a defined level)
R3 pull-up	GPIO8 → 3.3V	unchanged — GPIO8 is genuinely a strap pin here
GPIO0	strapping/boot use	now a spare GPIO, broken out but otherwise unused

This matches the same reset/boot convention Espressif uses on its own ESP32-C6-DevKitC and the Seeed XIAO ESP32C6.

2.2 Recommended addition: EN debounce/timing capacitor

Espressif's ESP32-C6 hardware design guidelines recommend a capacitor on EN for clean power-up reset timing (typical ~1μF, 100nF also common). This wasn't in your original list, so I have **not** added it to the schematic or BOM — flagging it as a recommended addition for you to accept or decline. If added, call it **C5, 1μF, EN → GND**.

Also worth noting: the same guidelines warn against adding a large capacitor directly on **GPIO9** — it can delay that pin reaching a valid strap level and cause an unwanted download-mode boot. Nothing here adds one there; just don't add decoupling to GPIO9 later.

2.3 J2 has no GND pin

The 4-pin MCU header (TXD/RXD/SRDY/MRDY) carries no ground reference. A UART link between two separate PCBs normally needs a common GND, or logic-level thresholds become dependent on however the two boards happen to be grounded elsewhere. Either confirm the mechanical/battery interface already ties both GNDs together and document that assumption, or add a 5th pin to J2 for GND.

2.4 Mechanical: 21.0 × 17.5mm is very tight

At the XIAO footprint size, 2.54mm pitch headers are almost certainly too large — a single 4-pin 2.54mm header alone is ~10mm long, and the board needs three connectors, two switches, an LDO, and eight passives. Move J2/J3 to 1.0-1.27mm pitch (or pogo-pin test pads for programming), use 0402 passives, and consider double-sided placement. This is a layout-stage decision but it changes the connector part numbers in the BOM, so resolve it before ordering.

3. Pin mapping (corrected)

U1 pin	Function	Direction	Connects to	Pull-up
VDDPST1, VDDPST2	Digital I/O power	in	+3.3V rail	C3, C4 (0.1μF each, local decoupling)
GND (multiple)	Ground	—	GND	—
EN (CHIP_PU)	Reset	in	R1 pull-up to 3.3V; SW1 to GND	R1, 10kΩ
GPIO8	Strap — ROM print control	in	R3 pull-up to 3.3V	R3, 10kΩ
GPIO9	Strap — boot mode select	in	R2 pull-up to 3.3V; SW2 to GND	R2, 10kΩ
GPIO0	Spare / unused	—	broken out, no connect	none
GPIO16	UART0 TXD	out	J2 pin1 (→ MCU RX), J3 pin2, TP4	—
GPIO7	UART0 RXD	in	J2 pin2 (← MCU TX), J3 pin3	—
GPIO3	SRDY (MCU→module wake)	in, active-low	J2 pin3	R4, 10kΩ
GPIO20	MRDY (module→MCU wake)	out, active-low	J2 pin4	R5, 10kΩ

GPIO16/GPIO7 are routed via the GPIO matrix rather than their IO_MUX default UART0 pins (GPIO16 default TX / GPIO17 default RX) — fine at 115200 baud, worth a firmware comment since it's non-default.

4. Net list

Net	Nodes
VBAT_IN	J1.1, D1 anode
VIN_PROT	D1 cathode, D2, C1+, U2.VIN
+3.3V	U2.VOUT, C2+, U1.VDDPST1, U1.VDDPST2, C3+, C4+, R1-R5 tops, J3.1, TP1
GND	J1.2, D2 cathode(-), C1-, U2.GND, C2-, U1.GND (all), C3-, C4-, SW1 far side, SW2 far side, J3.4, TP2
EN	U1.EN, R1 bottom, SW1 near side, TP3
GPIO8	U1.GPIO8, R3 bottom
GPIO9 / BOOT	U1.GPIO9, R2 bottom, SW2 near side
GPIO0	U1.GPIO0 (unconnected elsewhere)
TXD	U1.GPIO16, J2.1, J3.2, TP4
RXD	U1.GPIO7, J2.2, J3.3
SRDY	U1.GPIO3, J2.3, R4 bottom
MRDY	U1.GPIO20, J2.4, R5 bottom

5. Bill of materials

Ref	Qty	Description	Value / part	Package	Example MPN	Notes
U1	1	Wi-Fi 6 / BLE5 / 802.15.4 module	ESP32-C6-MINI-1-N8	SMD module, castellated	ESP32-C6-MINI-1-N8	8MB flash, PCB antenna
U2	1	LDO regulator, 3.3V	SGM2040-3.3 (or equiv.)	SOT23-5	SGM2040-3.3XN5G/TR	≥500mA, Iq<10μA. MCP1700/AP2112/XC6206 are similar alternates
D1	1	Schottky diode, reverse-polarity protection	SS12 (1A, 20V)	SMA / DO-214AC	SS12-E3/61T	Confirm Vf drop is acceptable at max load current
D2	1	Bidirectional TVS, ESD/transient protection	SMBJ5.0A	SMB / DO-214AA	SMBJ5.0A-13-F	Clamps VIN_PROT node to GND
C1	1	Bulk input capacitor	22μF X5R 10V (10-47μF range)	0805	GRM21BR61A226ME4 4L	

Ref	Qty	Description	Value / part	Package	Example MPN	Notes
C2	1	Bulk output capacitor	15 μ F X5R 6.3V (10-22 μ F range)	0805	GRM21BR60J156ME3 9L	
C3, C4	2	Decoupling, near U1 power pins	0.1 μ F X7R 16V	0402	GRM155R71C104KA8 8D	
R1-R5	5	Pull-up resistors	10k Ω , 1%	0402	RC0402FR-0710KL	R1 EN, R2 GPIO9, R3 GPIO8, R4 SRDY, R5 MRDY
J1	1	Battery connector, 2-pin	JST-PH 2.0mm or equiv.	THT/SMD, 2-pin	B2B-PH-K-S(LF)(SN)	Confirm mates with pack's existing connector
J2	1	Header to doorlock MCU, 4-pin	1.0-1.27mm pitch recommended	SMD header or pogo pad	—	2.54mm as originally specified likely won't fit the outline
J3	1	Programming header, 4-pin	1.0-1.27mm pitch recommended	SMD header or test pad	—	Same footprint caveat as J2
SW1	1	Reset button	Momentary tactile, low profile	SMD 3x3mm or smaller	TS-1187A-B-A-B	Wired to EN, not GPIO9
SW2	1	Boot/download button	Momentary tactile, low profile	SMD 3x3mm or smaller	TS-1187A-B-A-B	Wired to GPIO9, not GPIO0
TP1-TP4	4	Test points	3.3V, GND, EN, TXD	1mm pad or via	—	
C5 (optional)	1	EN timing/debounce cap — not yet in schematic	1 μ F (or 100nF)	0402/0603	—	See §2.2 — add only if confirmed

6. Power budget (sanity check)

- ESP32-C6 active Wi-Fi TX peak current: roughly 300-450mA class — confirm the exact figure against the module datasheet for your specific silicon revision before finalizing D1/U2 current ratings.
- SGM2040's 500mA and SS12's 1A ratings both clear that peak with headroom; D1's forward drop (~0.3-0.5V typical for a 1A Schottky at these currents) subtracts directly from the LDO's input headroom, so re-check U2's dropout spec at minimum battery voltage (3.6V) once the exact LDO part is chosen.
- Idle/TWT sleep current is where the “1.4+ years on 2400mAh” figure lives or dies — dominated by the LDO's quiescent current (SGM2040 spec'd sub-10 μ A) and the chip's light-sleep draw, not by peak TX. Worth a bench measurement before it goes on packaging.

7. Programming & bring-up procedure

- 1 Connect J3 (3.3V/TXD/RXD/GND) to a USB-UART adapter. Leave J2 disconnected from the doorlock MCU during programming, since J2 and J3 share the TXD/RXD net (GPIO16/GPIO7) — an active MCU on J2 could contend with the programmer.
- 2 Hold SW2 (GPIO9/BOOT) low, briefly pulse SW1 (EN/Reset), release SW2 — standard ESP32 download-mode entry.

- 3 Flash firmware via esptool or your toolchain of choice at the UART0 pins.
- 4 Release, reset via SW1, confirm normal boot (GPIO9 is pulled up by R2, so it defaults to SPI/normal boot when the button isn't held).
- 5 Reconnect J2 to the doorlock MCU for Keep-Alive integration testing; verify SRDY/MRDY handshake timing against the Tuya UART protocol spec before shipping.

8. Open items for next revision

- Decide on C5 (EN debounce cap) — see §2.2
- Resolve J2 GND reference — see §2.3
- Finalize J2/J3 connector pitch once placement is attempted at 21.0×17.5mm — see §2.4
- Confirm exact LDO part and re-check dropout margin at 3.6V / 500mA
- Confirm SRDY/MRDY timing against the specific Tuya Keep-Alive protocol revision targeted